

Complete IoT Developer Roadmap

Objective

Become a professional IoT Developer by mastering embedded systems, networking, cloud computing, and AI integration.

Phase 1: Programming

C
C++
Python

Phase 2: Electronics

Basic Electronics
Digital Electronics
Analog Electronics

Phase 3: Microcontrollers

Arduino

ESP32

STM32

Phase 4: Single Board Computers

Raspberry Pi

NVIDIA Jetson

Phase 5: Communication Protocols

UART

SPI

I2C

CAN

RS485

USB

Phase 6: Wireless Communication

Wi-Fi

Bluetooth

BLE

Zigbee

LoRa

NB-IoT

LTE-M

Phase 7: Networking

TCP/IP
MQTT
HTTP
HTTPS
WebSocket

Phase 8: Sensors

Temperature Sensors
Humidity Sensors
Gas Sensors
IMU
GPS
LiDAR
Camera Modules

Phase 9: Cloud Platforms

AWS IoT
Azure IoT
Google Cloud IoT

Phase 10: AI + IoT

TinyML
TensorFlow Lite

Edge AI
Computer Vision

Projects

Smart Home
Smart Agriculture
Weather Station
Industrial Monitoring
AI Surveillance System
Smart Energy Meter